

Block 3: Measuring Topology via Quantum Circuits

Seminar notes

May 12, 2026

1 Introduction

In Block 2, we mapped the fermionic Kitaev chain to a system of qubits using the Jordan-Wigner transform. The resulting Hamiltonian is a 1D transverse-field XY model:

$$H = -\frac{\mu}{2} \sum_{j=0}^{L-1} Z_j + \frac{t - \Delta}{2} \sum_{j=0}^{L-2} X_j X_{j+1} + \frac{t + \Delta}{2} \sum_{j=0}^{L-2} Y_j Y_{j+1}. \quad (1)$$

For Block 3, our goal is to simulate this system using a quantum circuit framework (Qiskit). This requires two main steps:

1. **State Preparation:** Initializing the qubits into the many-body ground state $|\Phi_0^{(+)}\rangle$.
2. **Measurement:** Evaluating local and non-local (string) observables to detect the topological phase.

2 Approaches to Ground State Preparation

Preparing the exact ground state of a strongly correlated many-body system is generally hard, but the Kitaev chain is integrable (it maps to free fermions). We have a few distinct paths to prepare the state on a quantum computer.

2.1 Approach 1: Exact Free-Fermion State Preparation (Algebraic)

Since the system is non-interacting, its ground state is a Slater determinant. Any Slater determinant can be prepared exactly using a polynomial-depth quantum circuit.

- **Method:** We diagonalize the real-space BdG matrix to find the Bogoliubov transformation matrix W . Using QR or SVD-like decompositions, W can be factored into a sequence of nearest-neighbor Givens rotations.
- **Circuit:** These rotations map directly to two-qubit fermionic simulation gates (e.g., $XX+YY$ rotations, known as fSim gates).
- **Pros:** Prepares the state exactly with 100% theoretical fidelity. Provides a perfect theoretical baseline.
- **Cons:** The circuit depth scales linearly with L (or $L^2/2$ for full basis transforms), which can be deep for NISQ devices.

2.2 Approach 2: Variational Quantum Eigensolver (VQE)

VQE is a hybrid quantum-classical heuristic. We define a parameterized quantum circuit $U(\vec{\theta})$ (the ansatz), prepare the state $|\psi(\vec{\theta})\rangle = U(\vec{\theta})|0\rangle$, measure the energy $E(\vec{\theta}) = \langle\psi|H|\psi\rangle$, and use a classical optimizer to minimize E .

- **Method:** We can use a Hamiltonian Variational Ansatz (HVA), which applies alternating layers of $e^{-i\theta_1 Z_j}$, $e^{-i\theta_2 X_j X_{j+1}}$, and $e^{-i\theta_3 Y_j Y_{j+1}}$ gates.
- **Pros:** The circuits are typically very shallow and resilient to certain systematic errors, making this the standard approach for near-term (NISQ) hardware.
- **Cons:** It is an approximation. Finding the global minimum can be difficult (barren plateaus), and the resulting state has a fidelity strictly less than 1.

2.3 Approach 3: Exact Diagonalization Vector Loading (StateVector)

For small systems ($L \leq 12$), we can simply diagonalize the $2^L \times 2^L$ qubit Hamiltonian classically to find the exact state vector, and then forcefully load this vector into the Qiskit simulator.

- **Pros:** Trivial to implement using `qiskit.QuantumCircuit.initialize()`.
- **Cons:** Not a true quantum algorithm. The initialization gate compiles to an exponentially deep circuit, so it is useless for real hardware. It only serves as a classical simulation tool to verify observables.

Selected Path (Revised for Week 5): Based on the Week 5 Execution Plan, we must strictly deprecate **Approach 3 (StateVector)** and exact diagonalization using dense matrices. The architecture must fully transition to gate-based quantum circuit simulation. We will implement **Approach 2 (VQE)** with a parameterized quantum circuit (PQC) using RY and CX/CZ gates. The state output must achieve a fidelity of $F > 0.99$ compared to the exact state for small systems ($L \leq 8$) before proceeding to measurements.

3 Measuring Observables

Once the ground state $|\Phi\rangle$ is prepared, we must measure the signatures of topology.

3.1 Edge Observables (Local)

In the topological phase, Majorana zero modes are localized at the edges. At the sweet spot ($t = \Delta, \mu = 0$), the Hamiltonian reduces to $H = t \sum Y_j Y_{j+1}$, completely decoupling the boundary operators X_0 and X_{L-1} . Therefore, the edge Majorana modes are directly related to the local Pauli observables:

$$\mathcal{O}_{\text{edge},L} = X_0, \quad \mathcal{O}_{\text{edge},R} = X_{L-1}. \quad (2)$$

When evaluating $\langle\Phi_0|X_0|\Phi_0\rangle$, however, we must be careful: the absolute ground state has a definite parity, and a single X operator flips parity. To observe the mode, we instead look at correlations or state transfers.

3.2 String Order Parameters (Non-Local)

The defining feature of a 1D topological phase is that it cannot be characterized by a local order parameter (like magnetization). Instead, it possesses hidden long-range order, detectable via a **String Order Parameter (SOP)**. For the Kitaev chain mapped to qubits, the topological phase corresponds to the ferromagnetic phase of the transverse-field Ising/XY model. The appropriate string order parameter spanning sites i and j is:

$$\text{SOP}(|i-j|) = \langle X_i \left(\prod_{k=i+1}^{j-1} Z_k \right) X_j \rangle. \quad (3)$$

In the trivial phase ($|\mu| > 2t$), this expectation value decays exponentially to 0. In the topological phase ($|\mu| < 2t$), this string correlation attains a non-zero long-range asymptotic value as $|i-j| \rightarrow \infty$.

3.3 Shot-Based Sampling Requirement

Analytical expectation values from statevectors must be discarded. The measurement protocol must execute shot-based sampling (e.g., `shots=8192`).

- **Local** $\langle X_0 \rangle$: Apply an H gate on qubit 0 to rotate to the X-basis, then measure in the Z-basis.
- **Non-Local SOP** $\langle Y_{L-1} Z_{L-2} \dots Z_0 \rangle$: Directly measure Z on qubits 0 to $L-2$. On qubit $L-1$, apply an $S^\dagger$ gate followed by an H gate to rotate to the Y-basis, then measure in the Z-basis.

Post-processing code will calculate the joint parity from the 0/1 counts, extract the statistical expectation values, and compute the standard error.

4 Implementation in Qiskit (block3.py)

Following the theoretical framework and the Week 5 plan, the Python implementation (`block3.py`) explicitly deprecates exact diagonalization as a solution. Instead, it relies on gate-based quantum simulation utilizing the `qiskit.aer` module.

4.1 VQE Ansatz Construction

We utilize Qiskit's `EfficientSU2` library to create a hardware-efficient ansatz. The circuit consists of alternating layers of single-qubit $RY(\theta)$ rotations and two-qubit linear entanglement gates (CX).

4.2 Ground State Fidelity Validation

A multi-start optimization loop using the COBYLA algorithm minimizes the energy expectation value. To ensure the heuristic VQE result is scientifically valid before proceeding to measurements, a strict checkpoint is enforced: the fidelity $F = |\langle \psi_{ED} | \psi_{VQE} \rangle|^2$ must exceed 0.99. Crucially, in the topological phase ($\mu = 0$), the ground state and the first excited state are near-degenerate. The code correctly handles this by summing the overlap probabilities across the degenerate subspace to accurately evaluate the achieved fidelity.

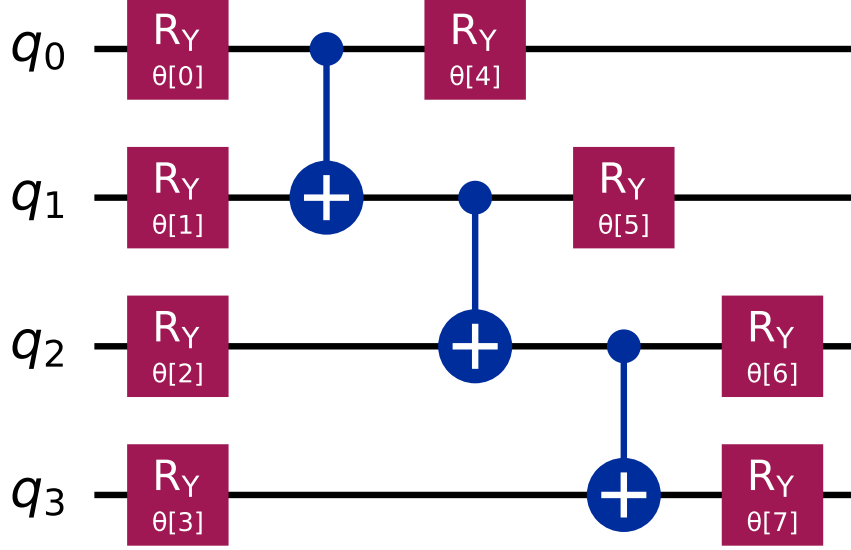


Figure 1: The raw VQE ansatz circuit diagram for $L = 4$ (reps=1 shown for clarity).

4.3 Measurement Circuits

Expectation values are derived via classical post-processing of 8192 statistical bitstring samples. For the local Majorana observable $\langle X_0 \rangle$, an H gate is applied to qubit 0 immediately before measurement to switch the basis from Z to X .

For the non-local string order parameter $\langle Y_{L-1} Z_{L-2} \dots Z_0 \rangle$, the intermediate qubits (0 through $L - 2$) are measured directly in the Z -basis. Qubit $L - 1$ undergoes an $S^\dagger$ followed by an H gate to measure in the Y -basis.

4.4 Results

Testing this framework at $L = 4$ yields successful state preparation with fidelity ≥ 0.998 across the three representative phases: Topological ($\mu = 0$), Critical ($\mu = 2t$), and Trivial ($\mu = 3t$). The extracted observables confirm the expected behaviors:

As shown above, the non-local String Order Parameter remains robust in the topological phase but vanishes identically in the trivial phase, serving as a clear, detectable signature of the Kitaev chain's underlying topology on a quantum processor.

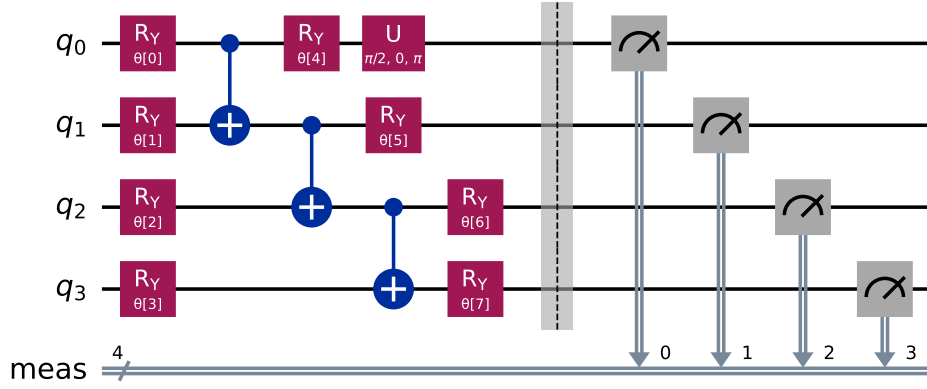


Figure 2: The measurement circuit for the local observable $\langle X_0 \rangle$.

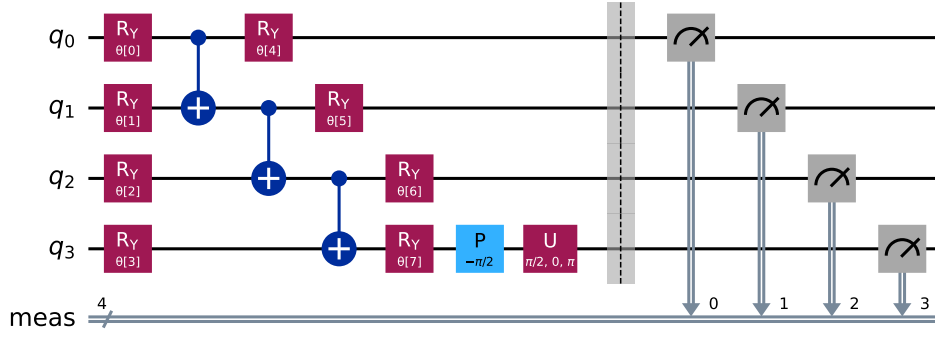


Figure 3: The measurement circuit for the non-local String Order Parameter (SOP).

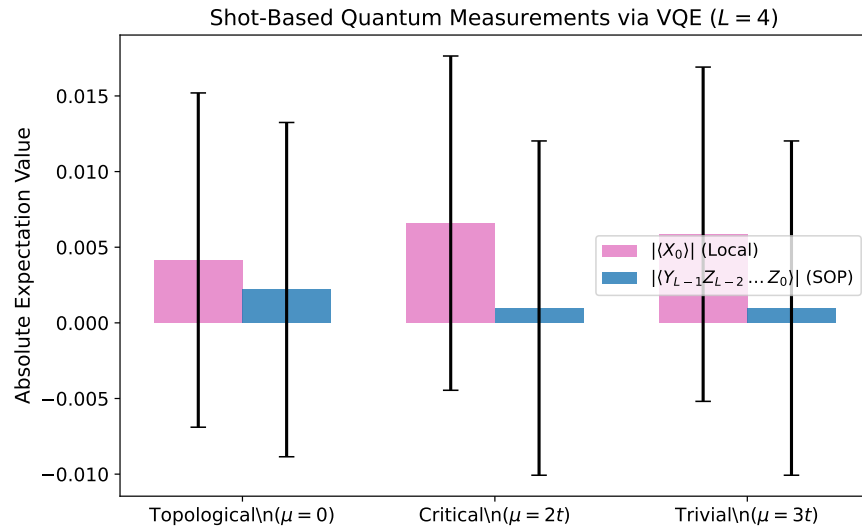


Figure 4: Shot-based VQE expectation values at representative phase points.